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Binance hack: what happened and how it could have been stopped

JP Creedon · Head of Digital Asset Technology, RBC Capital Markets · October 30, 2026 · Public sources; Binance's full investigation is not yet published

BOTTOM LINE

- **What:** on 29 October, attackers used a hacked engineer's laptop to get inside Binance and make its own payment-signing system approve their withdrawals. No secret keys were stolen, according to Binance.
- **Impact:** about **$6.3 billion** left Binance wallets on 10 networks in roughly 20 minutes, the largest crypto theft on record Estimates $6.3-6.4B by source. About $2.7-3 billion has been frozen; Binance says it will cover all customer losses.
- **Lesson:** the weak point was not the vault. It was what the signing system was willing to approve, who could ask it, and how quickly the outflow was stopped.

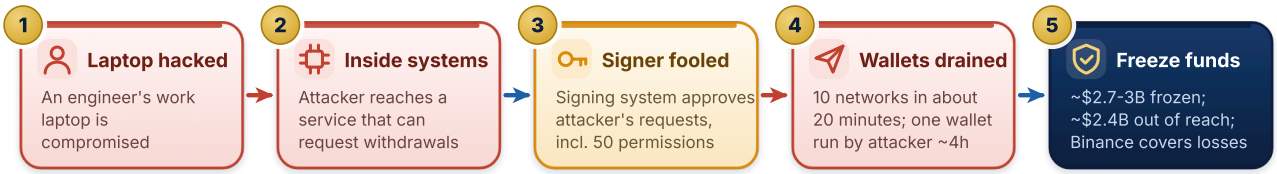
~$6.3B
taken (Binance estimate; analysts $6.36-6.4B)

10
blockchain networks hit

~20 min
for most of the outflow

~$2.7-3B
frozen so far; ~$2.4B out of reach

What happened



The attack in five steps. Steps 1-2 are Binance's initial finding; steps 3-5 come from blockchain-analytics firms tracing the funds. Most of the money moved in about 20 minutes on the evening of 29 October.

PRE-SIGNED PERMISSIONS, IN ONE SENTENCE

A pre-signed permission is an approval signed in advance that lets whoever holds it act on a wallet later, like a signed blank authority letter.

Here the attacker had Binance's own system sign 50 of them at once, so that one would still be valid at the moment of use; that one handed a large wallet to the attacker's program.

Where things stand

Binance

Withdrawals paused, with a target to resume in about 45 hours; deposits open. Binance says it will cover all losses and has set aside more than $6.3B.

Investigation

Binance's full technical report is still to come. It says it found "additional points of breach" it has not yet described.

Who did it

Analytics firms TRM and Chainalysis name North Korea's Lazarus group with high confidence Not officially confirmed.

HOW IT COULD HAVE BEEN STOPPED

One control per step

Our view, not Binance's findings. Any one of the first three controls holding would likely have stopped the attack; the last two would have shrunk the loss.

Step	Control that could have been in place	Effect
1 **Laptop hacked**	Engineers do sensitive work only on locked-down, disposable machines that cannot reach live systems	PREVENT
2 **Inside systems**	Access to live systems only with approval, for a limited time, and recorded; the signing system accepts requests only from verified services	PREVENT DETECT
3 **Signer fooled**	The signing system checks the actual payment itself (who, how much, why) and refuses anything unclear, such as a batch of open-ended permissions; a second, independent approver confirms each request against the original client instruction	PREVENT
4 **Wallets drained**	Limits on how much each wallet can send and to whom; keep only small balances in online wallets; an automatic alarm that halts withdrawals within minutes	DETECT LIMIT
5 **Freeze funds**	A rehearsed plan to cut off the attacker's access in minutes (it took about 3 hours 50 minutes here) and pre-arranged contacts with stablecoin issuers to freeze funds	LIMIT

PREVENT stops the step · DETECT raises the alarm · LIMIT caps the loss

What this means for us

1. **Make signing systems check what they sign.** Payments and permissions should be approved only when the system itself can read who, how much and why, with an independent second approver for anything unusual.
2. **Cap what any single failure can lose.** Per-wallet limits, small online balances topped up from offline storage, and an automatic stop that triggers in minutes, rehearsed in advance.
3. **Treat engineer access as a custody risk.** Locked-down machines and approval-only access to live systems; run this exact scenario as a tabletop exercise with Global Risk Management and Cyber.

General controls drawn from this incident; not an assessment of any RBC system.

Sources (full links in the detailed report, "Binance Incident Update"): Binance Security Incident Update (30 Oct 2026); TRM Labs; 808bits; BleepingComputer; CNBC; BBC. Figures differ by source and pricing time. Binance had not published its technical post-mortem as of October 30, 2026, ~10:45 AM ET; withdrawals were paused with a target to resume in about 45 hours.